

VORTEXA

Pre-Development Project Requirement Document

Date: 13 July 2026

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Executive Summary

The proposed platform is an enterprise-grade AI-powered Web3 investment ecosystem built around USDT, decentralized finance (DeFi), AI-based arbitrage trading, staking and blockchain transparency. Users invest using USDT while the platform automates market analysis, trading and risk protection. The objective is to provide a secure, transparent and intelligent investment environment where trading decisions are automated, user assets are protected during market volatility and all transactions remain verifiable on the blockchain.

Project Vision

Create a scalable DeFi platform that combines artificial intelligence, automated arbitrage, staking, blockchain verification and a reward ecosystem into a single unified solution.

Business Objectives

- USDT-based investment lifecycle.
- Automated AI trading.
- Capital protection during market crashes.
- Passive earning through staking.
- Transparent blockchain records.
- Centralized monitoring with admin dashboard.

Core Functional Requirements

- Secure authentication.
- Wallet connection.

- USDT deposits & withdrawals.
- Portfolio dashboard.
- AI arbitrage engine.
- Staking module.
- APR/APY display.
- Referral & reward system.
- Blockchain transaction history.
- Notification system.

AI Powered USDT Investment

Objective: Enable users to invest only through USDT.

Workflow: Wallet Connect → Deposit USDT → Blockchain Verification → AI Investment Allocation → Earnings Update → Withdrawal Request → Transaction Confirmation.

Expected Output: Secure, transparent and traceable investment lifecycle.

AI Arbitrage Trading Engine

Objective: Automatically identify profitable price differences across supported exchanges.

Functional Requirement: Continuously monitor market prices, detect arbitrage opportunities and execute trades without manual intervention.

Workflow: Market Scan → Opportunity Detection → AI Validation → Trade Execution → Profit Recording → Dashboard Update.

Expected Output: Continuous automated trading with minimal manual effort.

AI Umbrella Rule

Objective: Protect user capital during abnormal market conditions.

Workflow: AI monitors volatility → detects high-risk conditions → pauses aggressive trading → converts exposed assets into USDT → resumes trading after market stabilization.

Expected Output: Improved capital protection and controlled risk exposure.

DeFi & Staking

AI allocates eligible funds into DeFi opportunities and staking pools. Users can monitor APR, APY, reward history, estimated earnings and staking performance from their dashboard.

Blockchain Transparency

Every deposit, withdrawal and investment-related operation should be recorded and verifiable using blockchain transaction hashes, ensuring transparency and auditability.

Compensation & Reward System

Support Direct Business Bonus, Multi-Level Distribution, Passive Flow Income, Performance Rewards and Appraisal Bonuses with automated calculations according to business rules.

User Workflow

Register/Login → Connect Wallet → Deposit USDT → AI Investment → Monitor Portfolio → View Earnings → Withdraw Funds.

Admin Workflow

Monitor Users → Verify Transactions → Supervise AI Activity → Manage Staking → Review Withdrawals → Manage Rewards → View Reports & Analytics.

Required Third-Party Integrations

- Blockchain Wallet Integration
- Blockchain RPC Provider
- Decentralized Exchange Integration
- Market Data Provider
- AI Processing Service
- Email & Notification Service
- Security & Authentication Service

Non-Functional Requirements

High Availability, Scalability, Performance, Secure Communication, Encryption, Audit Logs, Data Integrity, Role-Based Access Control and Reliable Monitoring.

Project Complexity Assessment

Complexity: Enterprise Grade

Architecture: Distributed Web3 Platform

Blockchain: Required

AI Engine: Required

Admin Panel: Required

Security Level: High

Scalability: High

Professional Note

If any feature, workflow, business rule, operational logic or integration has been omitted or requires modification, kindly inform the development team before final approval of this document. This will ensure accurate project estimation, planning and implementation according to the client's business requirements.

Artificial Intelligence (AI) Engine

Objective: The AI Engine serves as the intelligent core of the platform, automating investment decisions, arbitrage trading, portfolio optimization and risk management.

Core Responsibilities:

- Monitor cryptocurrency markets continuously.
- Analyze price movements and volatility.
- Detect profitable arbitrage opportunities.
- Execute automated trading strategies.
- Optimize portfolio allocation.
- Protect user funds during abnormal market conditions.

AI Decision Workflow:

Market Analysis → Opportunity Detection → Risk Validation → Automated Trade Execution → Portfolio Update → Blockchain Transaction Recording.

Risk Management (Umbrella Rule):

During extreme volatility the AI should pause high-risk trading, convert exposed assets into USDT when required and resume automated trading after market stabilization.

AI Monitoring Dashboard:

Current AI Status, Trading Activity, Portfolio Performance, Risk Level, Total Executed Trades, Daily Performance and Investment Summary.

Expected Outcome: Deliver an intelligent, secure and transparent AI-driven investment ecosystem with automated trading, advanced market analysis and enhanced capital protection.

Project Success Criteria

The project will be considered successfully delivered when the following objectives are achieved:

- AI performs automated market analysis and arbitrage trading.
- Users can securely invest and withdraw using USDT.
- Blockchain transaction verification operates successfully.
- AI Risk Management (Umbrella Rule) functions correctly.
- DeFi and staking modules operate efficiently.
- User and Admin dashboards function as expected.

Future Enhancements (Optional)

- Mobile Application (Android & iOS)
- Additional Blockchain Network Support
- Advanced AI Prediction Models
- Multi-Language Support
- Advanced Portfolio Analytics

Final Statement

This document represents the current business requirements for the proposed AI-powered DeFi investment platform. Any additional features, business rules, workflows or integrations identified during future discussions should be reviewed and incorporated before the final Software Requirement Specification (SRS) and development phase begins.